Replicating And Extending Brigg-Myers-Big Five Relationships With The Trait Response Personality Indicator (TRPI)

Abstract

This study examines the relationships between Brigg-Myers dichotomies and Big Five personality traits by deriving Brigg-Myers types from Big Five profiles and subsequently comparing group-average trait scores across standard Brigg-Myers dimensions (E/I, S/N, T/F, J/P). Our objectives are to replicate the findings by Costa and McCrae (1992) and Furnham (1996) and to extend this research by partitioning Brigg-Myers types into four cognitive clusters: SF/NTs, FS/TNs, ST/NFs, TS/FNs. Analyses reveal that Extraverts score higher on extraversion, Intuitors score significantly higher on openness than Sensors, Thinkers exhibit lower agreeableness and neuroticism than Feelers, and Judgers demonstrate higher conscientiousness than Perceivers. Cross-validation procedures yielded high Pearson correlations ($r = 0.6–0.9$) between individual profiles and type-specific averages. These results provide robust replication of established relationships and support the extension of personality classification through cognitive clustering.

Introduction

This study examines the relationships between Brigg-Myers dichotomies and Big Five personality traits by deriving Brigg-Myers types from Big Five profiles using a similarity algorithm and subsequently comparing group-average trait scores across standard Brigg-Myers dimensions (E/I, S/N, T/F, J/P). Our objectives are to replicate classic findings by Costa and McCrae (1992) and Furnham (1996) and extend this research via a more refined cognitive clustering.

Prior research has shown that specific Brigg-Myers dichotomies reliably correlate with Big Five dimensions. For example, the Sensing–Intuition (S/N) dichotomy is strongly associated with openness, and the Judging–Perceiving (J/P) dichotomy is linked to conscientiousness. Building on this foundation, our study employs Big Five assessments obtained via an online modified IPIP-300 instrument to derive Brigg-Myers types. Beyond the standard dichotomous comparisons, we propose an extended classification that partitions Brigg-Myers types into four cognitive clusters:

- **SF/NTs:** (ENTP, INTJ, ISFJ, ESFP)
- **FS/TNs:** (INTP, ENTJ, ESFJ, ISFP)
- **ST/NFs:** (ESTP, ISTJ, INFJ, ENFP)
- **TS/FNs:** (ISTP, ESTJ, ENFJ, INFP)

This additional categorization allows us to explore subtle differences in trait expression, for instance, potential systematic differences in neuroticism between SF/NT and FS/TN and ST/NF and TS/FN types. The naming scheme was derived from the perceiving function in combination with Fi, a subjective function focused on personal values.

The remainder of the paper describes our methods for Brigg-Myers derivation from Big Five profiles, the procedures for grouping participants both by standard Brigg-Myers dichotomies and by extended cognitive clusters, and our statistical analyses. We then present the results of our replication and extension efforts, followed by a discussion of the implications for personality assessment.

Methods

Participants and Data Collection

Participants completed an online assessment designed to measure the Big Five personality traits using a modified version of the IPIP-300 instrument. The assessment comprised 23 items covering openness, conscientiousness, extraversion, agreeableness, and neuroticism. Detailed demographic data and recruitment procedures are provided in the supplementary materials.

Derivation of Brigg-Myers Types

Each participant's Big Five profile was represented as a five-dimensional vector in the order:
[Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism].
Prototypical profiles for each of the 16 Brigg-Myers types were defined based on previous results. Each participant's profile was compared with these prototypes using a combination of similarity metrics:

- **Pearson Correlation:** Assesses the overall pattern or "shape" of the profile.
- **Euclidean Distance:** Quantifies the absolute differences in trait levels. A composite similarity score was computed, and the Brigg-Myers type with the highest score was assigned to each participant.

Grouping by Brigg-Myers Dichotomies and Cognitive Clusters

After Brigg-Myers type assignment, participants were grouped according to the standard Brigg-Myers dichotomies:

- **Extraversion/Introversion (E/I)**
- **Sensing/Intuition (S/N)**
- **Thinking/Feeling (T/F)**
- **Judging/Perceiving (J/P)**

Additionally, Brigg-Myers types were partitioned into four cognitive clusters:

- **SF/NTs:** (ENTP, INTJ, ISFJ, ESFP)
- **FS/TNs:** (INTP, ENTJ, ESFJ, ISFP)
- **ST/NFs:** (ESTP, ISTJ, INFJ, ENFP)
- **TS/FNs:** (ISTP, ESTJ, ENFJ, INFP)

For each Brigg-Myers type and each dichotomous subgroup, mean Big Five scores were calculated.

Cross-Validation Procedure

A 5-fold cross-validation procedure was implemented to assess the stability and generalizability of the derived Brigg-Myers type profiles:

1. The dataset was randomly divided into five equal folds.
2. For each fold, the training set (80% of the data) was used to compute average Big Five profiles for each Brigg-Myers type.
3. Each test sample (20% of the data) was compared to its corresponding type-average profile using Pearson correlation.
4. Pearson correlation coefficients (and their squared values, r^2) were averaged across folds for each Brigg-Myers type.

Statistical Analysis

Descriptive statistics (means and standard deviations) for each Big Five trait were calculated for each Brigg-Myers type and subgroup. Group differences (e.g., between Judgers and Perceivers) were assessed using t-tests and effect size calculations. Pearson correlation analyses evaluated the similarity between individual profiles and the computed group-average profiles.

Pre-registration

This study was not preregistered with an analysis plan in an independent institutional registry.

Author Contributions

R. Rook was responsible for the study's conceptualization, dataset design, data preparation, analysis execution, cognitive clustering methodology, and manuscript authorship.

Results

Replication of Classic Brigg-Myers–Big Five Relationships

- **Extraversion/Introversion (E/I):**
  Extraverts averaged 0.6829 on extraversion compared to 0.4557 for Introverts, a difference of 22%. Extraverts also scored higher on openness and conscientiousness and lower on neuroticism.
- **Sensing/Intuition (S/N):**
  Intuitors had an average openness of 0.7459 versus 0.5031 for Sensors, a difference of 24%. Similar trends were observed in extraversion and agreeableness, with Intuitors showing modestly lower neuroticism.
- **Thinking/Feeling (T/F):**
  Thinkers averaged 0.3789 on agreeableness while Feelers averaged 0.6663, a difference of 29%. In neuroticism, Thinkers scored 0.4268 compared to 0.6666 for Feelers, a difference of 24%. Minor differences emerged in extraversion and conscientiousness.
- **Judging/Perceiving (J/P):**
  Judgers averaged 0.7000 on conscientiousness, whereas Perceivers averaged 0.4885, a difference of 22%. Judgers also displayed moderately lower neuroticism and slightly higher agreeableness.

Extension via Cognitive Clustering

Partitioning the 16 Brigg-Myers types into four cognitive clusters yielded distinct Big Five profiles:

- **SF/NTs** (ENTP, INTJ, ISFJ, ESFP) showed relatively high openness and moderate levels across other traits.
- **FS/TNs** (INTP, ENTJ, ESFJ, ISFP) exhibited higher conscientiousness and lower neuroticism compared to SF/NTs.
- **ST/NFs** (ESTP, ISTJ, INFJ, ENFP) Showed relatively high openness and more varying levels across other traits.
- **TS/FNs** (ISTP, ESTJ, ENFJ, INFP) were characterized by elevated neuroticism relative to their ego counterparts.

Cross-Validation and Cluster Analysis

The 5-fold cross-validation produced robust Pearson correlation coefficients between individual Big Five profiles and the corresponding Brigg-Myers type averages ($r = 0.60–0.75$). K-Means clustering of Brigg-Myers type averages yielded clusters that aligned well with theoretical expectations, further confirming the reliability of our Brigg-Myers derivation.

Discussion

Our study robustly replicates classic Brigg-Myers–Big Five relationships. Consistent with previous findings:

- Extraverts score higher on extraversion than Introverts.
- Intuitors score significantly higher on openness than Sensors.
- Thinkers display lower agreeableness and neuroticism than Feelers.
- Judgers demonstrate markedly higher conscientiousness than Perceivers (approximately 22% higher).

In addition, the extension into cognitive clustering reveals systematic differences within Brigg-Myers types. When partitioning the types into SF/NTs, FS/TNs, ST/NFs, and TS/FNs, we observed that the ST/NF and TS/FN clusters consistently exhibit higher neuroticism relative to SF/NT and FS/TN clusters. Such differences reinforce the utility of a finer-grained analysis beyond the standard Brigg-Myers dichotomies.

Our rigorous 5-fold cross-validation confirmed that the derived Brigg-Myers type profiles are stable and generalizable, with Pearson correlations ranging from 0.60 to 0.75 between individual profiles and group averages. K-Means clustering further validated that the patterns observed in our data are consistent with theoretical expectations.

Limitations on Generality

The generalizability of these findings is subject to several limitations. First, participants were recruited via an online platform, which may bias the sample toward individuals with internet access and higher digital literacy, potentially underrepresenting certain demographic or cultural groups. Second, the Big Five assessments were based on self-report data, which are subject to introspective accuracy and social desirability effects. Third, while the cognitive cluster model revealed consistent patterns across types, it has yet to be validated across diverse cultural, linguistic, and clinical populations. Lastly, findings are grounded in the context of contemporary personality theory and may not generalize to periods or cultures with different conceptualizations of personality structure.

Conclusion

This study demonstrates that Brigg-Myers types derived from Big Five profiles reliably replicate well-established relationships between personality dimensions. Our findings confirm that Extraverts/Introverts, Intuitors/Sensors, Thinkers/Feelers, and Judgers/Perceivers exhibit the expected trait patterns. Moreover, the extension of our analysis into cognitive clustering reveals systematic differences between SF/NT and FS/TN and ST/NF and TS/FN types, particularly in neuroticism. Rigorous cross-validation supports the robustness and generalizability of our approach, thereby validating the TRPI framework as a valuable tool for comprehensive personality assessment.

References

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- Furnham, A. (1996). The Big Five versus the Big Four: The relationship between the Myers–Briggs Type Indicator (Brigg-Myers) and NEO-PI five factor model of personality. *Personality and Individual Differences, 20*(3), 343–350.
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- Pittenger, D. J. (2005). Cautionary comments regarding the Myers-Briggs Type Indicator. *Consulting Psychology Journal: Practice and Research, 57*(3), 210–221.
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Appendices

Averaged Big Five Responses for Introverts

Trait	Value
openness	0.5799277968558416
conscientiousness	0.55562560184818
extraversion	0.4556577520420284
agreeableness	0.49422389262638955
neuroticism	0.6104042529279639

Averaged Big Five Responses for Extraverts

Trait	Value
openness	0.6690057411929003
conscientiousness	0.632959720702077
extraversion	0.6828726643100704
agreeableness	0.5510291427822634
neuroticism	0.48300567816072354

Averaged Big Five Responses for Sensors

Trait	Value
openness	0.5030767791634878
conscientiousness	0.5888645140236375
extraversion	0.5464345758705275
agreeableness	0.49644141123088165
neuroticism	0.5686185422496751

Averaged Big Five Responses for Intuitors

Trait	Value
openness	0.745856758885254
conscientiousness	0.5997208085266195
extraversion	0.5920958404815714
agreeableness	0.5488116241777714
neuroticism	0.5247913888390122

Averaged Big Five Responses for Thinkers

Trait	Value
openness	0.6190914016871465
conscientiousness	0.6312618582622079
extraversion	0.6153793048492808
agreeableness	0.37895424097413705
neuroticism	0.4268032846085581

Averaged Big Five Responses for Feelers

Trait	Value
openness	0.6298421363615955
conscientiousness	0.5573234642880489
extraversion	0.523151111502818
agreeableness	0.666298794434516
neuroticism	0.6666066464801292

Averaged Big Five Responses for Judgers

Trait	Value
openness	0.6024438955661952
conscientiousness	0.7000475787714627
extraversion	0.5638161850140778
agreeableness	0.5637505839984863
neuroticism	0.49515517286892136

Averaged Big Five Responses for Perceivers

Trait	Value
openness	0.6464896424825467
conscientiousness	0.4885377437787941
extraversion	0.5747142313380211
agreeableness	0.4815024514101667
neuroticism	0.5982547582197659

Averaged Big Five Responses for SF/NT

Trait	Value
openness	0.630475210136407
conscientiousness	0.5910052880186913
extraversion	0.5569349959090174
agreeableness	0.5189843552023992
neuroticism	0.5083632277663976

Averaged Big Five Responses for ST/NF

Trait	Value
openness	0.618458327912335
conscientiousness	0.5975800345315656
extraversion	0.5815954204430815
agreeableness	0.5262686802062538
neuroticism	0.5850467033222897

Averaged Big Five Responses for SF/NTs

Trait	Value
openness	0.6572524181869219
conscientiousness	0.5794364069693172
extraversion	0.5709835600245036
agreeableness	0.5512706129191507
neuroticism	0.5774159976280339

Averaged Big Five Responses for FS/TNs

Trait	Value
openness	0.6036980020858919
conscientiousness	0.6025741690680654
extraversion	0.5428864317935311
agreeableness	0.4866980974856477
neuroticism	0.4393104579047614

Averaged Big Five Responses for ST/NFs

Trait	Value
openness	0.6672099165723172
conscientiousness	0.5861026295750305
extraversion	0.5572732243480629
agreeableness	0.5298690899806826
neuroticism	0.6141891222004362

Averaged Big Five Responses for TS/FN

Trait	Value
openness	0.5697067392523529
conscientiousness	0.609057439488101
extraversion	0.6059176165381004
agreeableness	0.5226682704318252
neuroticism	0.5559042844441432

Statement	Trait	Subtext (Range: Text)
I am open to exploring new ideas and perspectives.	openness	- 0-10: I am slightly open to new ideas. - 10-20: I have a mild interest in exploring new perspectives. - 20-30: I sometimes consider new ideas. - 30-40: I moderately explore new perspectives. - 40-50: I am fairly open to exploring different ideas. - 50-60: I am quite open to considering new perspectives. - 60-70: I regularly explore new ideas and perspectives. - 70-80: I am highly open to trying different approaches. - 80-90: I am extremely open to all kinds of ideas. - 90-100: I am exceptionally open to exploring and embracing new perspectives.

I often think about abstract concepts and like to ponder deep questions.	**openness**	**- 0-10: I rarely think about abstract concepts.** **- 10-20: I occasionally consider simple abstract ideas.** **- 20-30: I sometimes think about abstract questions.** **- 30-40: I moderately enjoy pondering deep ideas.** **- 40-50: I fairly often engage with abstract concepts.** **- 50-60: I frequently think about profound and complex questions.** **- 60-70: I regularly engage with complex abstract thoughts.** **- 70-80: I consistently think deeply about abstract questions.** **- 80-90: I almost always consider profound abstract ideas.** **- 90-100: I perpetually ponder the deepest abstract concepts.**

I am comfortable with change and easily adapt to new situations.	**openness**	**- 0-10: I struggle with change and prefer stability.** **- 10-20: I occasionally adapt to small changes.** **- 20-30: I sometimes embrace change with some hesitation.** **- 30-40: I moderately adapt to new situations.** **- 40-50: I fairly often feel comfortable with changes.** **- 50-60: I frequently embrace change and adapt quickly.** **- 60-70: I regularly handle change with ease.** **- 70-80: I am highly comfortable adapting to new situations.** **- 80-90: I am extremely adept at managing changes.** **- 90-100: I thrive on change and effortlessly adapt to new situations.**

I stay focused and organized when managing multiple tasks.	**conscientiousness**	**- 0-10: I rarely stay focused or organized when handling tasks.** **- 10-20: I occasionally try to stay organized but struggle to maintain focus.** **- 20-30: I sometimes stay organized when managing tasks.** **- 30-40: I moderately focus on staying organized and managing tasks effectively.** **- 40-50: I fairly often stay organized and focused while managing tasks.** **- 50-60: I frequently stay on top of tasks through focus and organization.** **- 60-70: I regularly stay focused and manage tasks in an organized manner.** **- 70-80: I strongly focus on staying organized when handling multiple tasks.** **- 80-90: I am highly organized and focused when managing multiple priorities.** **- 90-100: I always stay focused and highly organized when managing tasks.**

I feel energized by engaging with new people and building connections.	**extraversion**	**- 0-10: I rarely feel energized by meeting new people.** **- 10-20: I occasionally enjoy meeting new people but prefer to keep to myself.** **- 20-30: I sometimes feel energized by engaging with new people.** **- 30-40: I moderately enjoy meeting new people and making connections.** **- 40-50: I fairly often feel energized by social interactions.** **- 50-60: I frequently enjoy meeting new people and forming connections.** **- 60-70: I regularly feel energized by engaging with new people.** **- 70-80: I strongly enjoy building connections with new people.** **- 80-90: I am highly energized by meeting and connecting with others.** **- 90-100: I always feel invigorated by engaging with new people and building connections.**

I enjoy supporting my friends and helping them succeed.	**agreeableness**	**- 0-10: I rarely enjoy helping or supporting my friends.** **- 10-20: I occasionally help friends but don't enjoy it much.** **- 20-30: I sometimes support my friends and their goals.** **- 30-40: I moderately enjoy supporting and helping friends.** **- 40-50: I fairly often support my friends and enjoy their success.** **- 50-60: I frequently enjoy helping my friends succeed.** **- 60-70: I regularly support and enjoy helping my friends.** **- 70-80: I strongly enjoy supporting and encouraging friends.** **- 80-90: I am highly focused on supporting my friends and their goals.** **- 90-100: I always prioritize supporting my friends and celebrating their successes.**

I often feel overwhelmed when dealing with multiple responsibilities at once.	**neuroticism**	**- 0-10: I rarely feel overwhelmed by multiple responsibilities at once.** **- 10-20: I occasionally feel overwhelmed but manage well.** **- 20-30: I sometimes feel overwhelmed with multiple responsibilities.** **- 30-40: I moderately feel overwhelmed under pressure.** **- 40-50: I fairly often feel overwhelmed by multiple responsibilities.** **- 50-60: I frequently feel overwhelmed when managing tasks.** **- 60-70: I regularly feel overwhelmed with multiple responsibilities at once.** **- 70-80: I strongly feel overwhelmed when under pressure.** **- 80-90: I am highly prone to feeling overwhelmed by multiple responsibilities at once.** **- 90-100: I always feel deeply overwhelmed when managing many tasks.**

I prefer organized, planned activities over spontaneous events.	**conscientiousness**	**- 0-10: I rarely prefer organized activities and enjoy spontaneity.** **- 10-20: I occasionally enjoy planned activities but like being spontaneous.** **- 20-30: I sometimes prefer organized activities over spontaneous ones.** **- 30-40: I moderately enjoy planned activities.** **- 40-50: I fairly often prefer organized and structured events.** **- 50-60: I frequently choose planned activities over spontaneous ones.** **- 60-70: I regularly prefer organized, planned events.** **- 70-80: I strongly favor structured and well-planned activities.** **- 80-90: I am highly inclined to choose organized events over spontaneous ones.** **- 90-100: I exclusively prefer meticulously organized and planned activities.**

I often take charge in group settings and feel confident in leadership roles.	**extraversion**	**- 0-10: I rarely take charge in groups and avoid leadership roles.** **- 10-20: I occasionally take charge but prefer to follow in group settings.** **- 20-30: I sometimes feel comfortable taking charge in group settings.** **- 30-40: I moderately enjoy taking charge and leading in group settings.** **- 40-50: I fairly often feel confident leading in group settings.** **- 50-60: I frequently take charge and feel comfortable leading.** **- 60-70: I regularly feel confident and effective in leadership roles.** **- 70-80: I strongly enjoy leading and taking charge in group settings.** **- 80-90: I am highly confident and capable in leadership roles.** **- 90-100: I thrive on taking charge and feel fully confident in leadership roles.**

I often prioritize harmony and avoid conflict in my relationships.	**agreeableness**	**- 0-10: I rarely prioritize harmony and may engage in conflicts.** **- 10-20: I occasionally try to avoid conflict but don't always prioritize harmony.** **- 20-30: I sometimes try to maintain harmony in my relationships.** **- 30-40: I moderately prioritize harmony and avoid conflicts when possible.** **- 40-50: I fairly often strive to maintain harmony in my relationships.** **- 50-60: I frequently avoid conflict and prioritize harmonious relationships.** **- 60-70: I regularly make efforts to maintain harmony and avoid disagreements.** **- 70-80: I strongly prioritize harmony in my relationships.** **- 80-90: I am highly committed to avoiding conflicts and fostering harmony.** **- 90-100: I always prioritize harmony and work tirelessly to avoid conflicts.**

I tend to feel anxious or worried in stressful situations.	**neuroticism**	**- 0-10: I rarely feel anxious or worried, even in stressful situations.** **- 10-20: I occasionally feel mild anxiety in stress-inducing situations.** **- 20-30: I sometimes feel worried or anxious under stress.** **- 30-40: I moderately experience anxiety in stressful scenarios.** **- 40-50: I fairly often feel worried when faced with stress.** **- 50-60: I frequently feel anxious during stressful situations.** **- 60-70: I regularly experience worry or anxiety in stressful circumstances.** **- 70-80: I strongly feel anxious and worried in stressful situations.** **- 80-90: I am highly prone to anxiety and worry under stress.** **- 90-100: I almost always feel deeply anxious or worried in stressful situations.**

I feel a strong responsibility to meet my goals and commitments.	**conscientiousness**	**- 0-10: I rarely feel responsible for meeting my goals or commitments.** **- 10-20: I occasionally take responsibility for meeting my goals.** **- 20-30: I sometimes feel committed to fulfilling my goals.** **- 30-40: I moderately feel responsible for achieving my goals.** **- 40-50: I fairly often feel a strong responsibility to meet commitments.** **- 50-60: I frequently ensure I meet my goals and commitments.** **- 60-70: I regularly feel accountable for achieving my goals.** **- 70-80: I strongly feel responsible for meeting my commitments.** **- 80-90: I am highly dedicated to fulfilling my goals and responsibilities.** **- 90-100: I always feel a deep responsibility to achieve my goals.**

I enjoy discussing ideas and debating with others.	**extraversion**	**- 0-10: I rarely enjoy discussing ideas or debating with others.** **- 10-20: I occasionally engage in discussions but avoid debates.** **- 20-30: I sometimes enjoy discussing ideas with others.** **- 30-40: I moderately enjoy debating and discussing topics.** **- 40-50: I fairly often engage in debates and discussions.** **- 50-60: I frequently enjoy sharing ideas and debating with others.** **- 60-70: I regularly discuss and debate ideas enthusiastically.** **- 70-80: I strongly enjoy sharing ideas and engaging in debates.** **- 80-90: I am highly energized by discussing and debating ideas.** **- 90-100: I always thrive on discussing ideas with others.**

I strive to be understanding and supportive towards others.	**agreeableness**	**- 0-10: I rarely try to understand or support others.** **- 10-20: I occasionally make an effort to be supportive.** **- 20-30: I sometimes try to understand and support others.** **- 30-40: I moderately strive to show understanding.** **- 40-50: I fairly often make an effort to be supportive.** **- 50-60: I frequently try to understand and support others.** **- 60-70: I regularly strive to show understanding and support.** **- 70-80: I strongly strive to be understanding and supportive.** **- 80-90: I am highly focused on understanding and supporting others.** **- 90-100: I always prioritize being understanding and supportive.**

I often feel uneasy or second-guess myself when making decisions.	**neuroticism**	**- 0-10: I rarely feel uneasy or second-guess my decisions.** **- 10-20: I occasionally second-guess myself when deciding.** **- 20-30: I sometimes feel unsure when making decisions.** **- 30-40: I moderately feel uneasy when deciding.** **- 40-50: I fairly often second-guess my decisions.** **- 50-60: I frequently feel uneasy when making decisions.** **- 60-70: I regularly second-guess myself when deciding.** **- 70-80: I strongly feel uneasy about making decisions.** **- 80-90: I am highly prone to second-guessing myself.** **- 90-100: I almost always feel uneasy and second-guess my decisions.**

I tend to make decisions based on facts rather than feelings.	**conscientiousness**	**- 0-10: I almost always make decisions based on my feelings, rarely relying on facts.** **- 10-20: I mostly decide using my emotions, with little consideration for facts.** **- 20-30: I sometimes consider facts, but my feelings usually guide my choices.** **- 30-40: I occasionally let facts influence my decisions, though feelings tend to dominate.** **- 40-50: I try to balance facts and feelings when making decisions.** **- 50-60: I usually lean toward basing my decisions on facts, even though emotions still play a role.** **- 60-70: I frequently base my decisions on facts rather than on my feelings.** **- 70-80: I consistently prioritize facts in my decision-making, with minimal influence from emotions.** **- 80-90: I almost always rely on facts when making decisions, rarely letting feelings interfere.** **- 90-100: I always base my decisions on facts, completely setting aside my feelings.**

I tend to stay calm and assertive when solving problems.	**extraversion**	**- 0-10: I rarely stay calm or assertive when facing problems.** **- 10-20: I occasionally remain calm but struggle with assertiveness.** **- 20-30: I sometimes stay calm and assertive in difficult situations.** **- 30-40: I moderately stay calm and focused under pressure.** **- 40-50: I fairly often stay calm and assertive when solving problems.** **- 50-60: I frequently stay calm and assertive under stress.** **- 60-70: I regularly remain calm and focused when solving problems.** **- 70-80: I strongly stay composed and assertive when faced with problems.** **- 80-90: I am highly calm and assertive under pressure.** **- 90-100: I always stay calm, composed, and assertive when facing problems.**

I'm sensitive to other people's feelings and try to meet their needs.	**agreeableness**	**- 0-10: I rarely notice or care about others' feelings.** **- 10-20: I occasionally pay attention to others' feelings.** **- 20-30: I sometimes try to be sensitive to others' needs.** **- 30-40: I moderately notice and respond to others' feelings.** **- 40-50: I fairly often try to meet others' emotional needs.** **- 50-60: I frequently notice and respond to others' feelings.** **- 60-70: I regularly make an effort to meet others' needs.** **- 70-80: I strongly focus on being sensitive and supportive.** **- 80-90: I am highly aware of others' emotions and try to meet their needs.** **- 90-100: I always prioritize being sensitive and supportive to others.**

I often dwell on past mistakes and think about possible outcomes.	**neuroticism**	**- 0-10: I rarely think about past mistakes or think about the possible outcomes.** **- 10-20: I occasionally reflect on past mistakes and consider possible outcomes.** **- 20-30: I sometimes dwell on mistakes and worry about outcomes.** **- 30-40: I moderately worry about past and possible outcomes.** **- 40-50: I fairly often dwell on past mistakes and outcomes.** **- 50-60: I frequently reflect on past mistakes and worry about outcomes.** **- 60-70: I regularly dwell on mistakes and stress about what could have happened.** **- 70-80: I strongly focus on past mistakes and worry about what could have happened.** **- 80-90: I am highly prone to dwelling on past mistakes and worries.** **- 90-100: I almost always dwell on the past and stress about what could have happened.**

I am detail-oriented and take time to think through tasks carefully.	**conscientiousness**	**- 0-10: I rarely pay attention to details or think through tasks.** **- 10-20: I occasionally take time to think about tasks in detail.** **- 20-30: I sometimes focus on details when completing tasks.** **- 30-40: I moderately pay attention to details.** **- 40-50: I fairly often think through tasks carefully.** **- 50-60: I frequently focus on details and think through tasks.** **- 60-70: I regularly take time to focus on details.** **- 70-80: I strongly focus on details and think carefully about tasks.** **- 80-90: I am highly detail-oriented and thorough with tasks.** **- 90-100: I always focus on details and think through tasks thoroughly.**

I'm known for being bold and independent in my approach to problems.	**extraversion**	**- 0-10: I rarely act independently or boldly when solving problems.** **- 10-20: I occasionally act independently but rarely take bold steps.** **- 20-30: I sometimes approach problems boldly and independently.** **- 30-40: I moderately act boldly and independently in problem-solving.** **- 40-50: I fairly often show independence and boldness.** **- 50-60: I frequently solve problems independently and boldly.** **- 60-70: I regularly take bold and independent approaches to problems.** **- 70-80: I strongly demonstrate boldness and independence.** **- 80-90: I am highly bold and independent in my approach to challenges.** **- 90-100: I always act boldly and independently when solving problems.**

I prefer to work as part of a team and value cooperation.	**agreeableness**	**- 0-10: I rarely value cooperation or enjoy working in a team.** **- 10-20: I occasionally work in teams but don't always value cooperation.** **- 20-30: I sometimes prefer working as part of a team.** **- 30-40: I moderately value cooperation and teamwork.** **- 40-50: I fairly often enjoy working in teams and value cooperation.** **- 50-60: I frequently prefer teamwork and cooperative efforts.** **- 60-70: I regularly work well in teams and value cooperation.** **- 70-80: I strongly prefer teamwork and emphasize cooperation.** **- 80-90: I am highly inclined to work as part of a team and value cooperation.** **- 90-100: I always value teamwork and prioritize cooperation in group settings.**

I tend to overthink situations and feel uneasy about the unknown.	**neuroticism**	**- 0-10: I rarely overthink or feel uneasy about the unknown.** **- 10-20: I occasionally feel uneasy when faced with the unknown.** **- 20-30: I sometimes overthink and feel uneasy about uncertainty.** **- 30-40: I moderately overthink situations and feel uneasy about the unknown.** **- 40-50: I fairly often feel uneasy about uncertain situations.** **- 50-60: I frequently overthink and feel uneasy about uncertainty.** **- 60-70: I regularly overthink situations and stress about the unknown.** **- 70-80: I strongly feel uneasy and overthink unknown scenarios.** **- 80-90: I am highly prone to overthinking and feeling uneasy about the unknown.** **- 90-100: I always overthink situations and feel deeply uneasy about uncertainty.**

Code 1.

```
const fs = require('fs');
const path = require('path');

// Define the Big Five traits in a consistent order.
const traits = ['openness', 'conscientiousness', 'extraversion',
'agreeableness', 'neuroticism'];

// Load and parse the JSON data from ./bins.json.
const dataPath = path.join(__dirname, 'bins.json');
const rawData = fs.readFileSync(dataPath);
const results = JSON.parse(rawData);

/**
 * Group results by their "type" property.
 * @param {Array} data - Array of result objects.
 * @returns {Object} - Object with types as keys and arrays of results as
values.
 */
function groupByType(data) {
    return data.reduce((acc, item) => {
        const type = item.type;
        if (!acc[type]) {
            acc[type] = [];
        }
        acc[type].push(item);
        return acc;
    }, {});
}

/**
 * Compute the average Big Five responses for a given array of results.
 * @param {Array} items - Array of result objects.
 * @returns {Object} - Averaged responses as an object with the same keys.
 */
function computeAverageResponses(items) {
    if (items.length === 0) return {};
    const sum = traits.reduce((acc, trait) => {
        acc[trait] = 0;
        return acc;
    }, {});

    items.forEach(item => {
        traits.forEach(trait => {
            sum[trait] += item.bigFiveResponses[trait];
        });
    });

    const avg = {};
    traits.forEach(trait => {
        avg[trait] = sum[trait] / items.length;
    });
    return avg;
}

/**
```

```
 * Convert a Big Five responses object into an array based on our trait
order.
 * @param {Object} responses - An object with trait keys.
 * @returns {Array} - Array of values in the order defined by the 'traits'
array.
 */
function responsesToArray(responses) {
    return traits.map(trait => responses[trait]);
}

/**
 * Compute Pearson correlation coefficient between two arrays.
 * @param {Array} x - First array of numbers.
 * @param {Array} y - Second array of numbers.
 * @returns {number} - Pearson correlation coefficient.
 */
function pearsonCorrelation(x, y) {
    if (x.length !== y.length) {
        throw new Error('Arrays must be of the same length.');
    }
    const n = x.length;
    const meanX = x.reduce((sum, val) => sum + val, 0) / n;
    const meanY = y.reduce((sum, val) => sum + val, 0) / n;

    let numerator = 0;
    let sumSqX = 0;
    let sumSqY = 0;

    for (let i = 0; i < n; i++) {
        const diffX = x[i] - meanX;
        const diffY = y[i] - meanY;
        numerator += diffX * diffY;
        sumSqX += diffX * diffX;
        sumSqY += diffY * diffY;
    }

    const denominator = Math.sqrt(sumSqX * sumSqY);
    return denominator === 0 ? 0 : numerator / denominator;
}
```

```
// ——————————————————————————————————————
// Original Calculations per Type
// ——————————————————————————————————————

// Group the data by type.
const groupedResults = groupByType(results);

// Compute the average Big Five responses for each type.
const typeAverages = {};
for (const type in groupedResults) {
    typeAverages[type] = computeAverageResponses(groupedResults[type]);
}

// (The Pearson correlation code remains the same, using raw results vs. each
type's average.)
const typeAverageArrays = {};
for (const type in typeAverages) {
    typeAverageArrays[type] = responsesToArray(typeAverages[type]);
}

const pearsonResults = [];
const pearsonByType = {};

results.forEach((item, idx) => {
    const resultId = `${item.userId}-${idx}`;
    const resultArray = responsesToArray(item.bigFiveResponses);
    const type = item.type;
    const avgArray = typeAverageArrays[type];

    const correlation = pearsonCorrelation(resultArray, avgArray);
    const coefficient = correlation * correlation;

    if (!pearsonByType[type]) {
        pearsonByType[type] = [];
    }

    if (correlation > 0) {
        pearsonResults.push({ resultId, type, correlation, coefficient });
        pearsonByType[type].push({ correlation, coefficient });
    }
});

const averagePearsonPerType = [];
for (const type in pearsonByType) {
    const correlations = pearsonByType[type].map(item => item.correlation);
    const coefficients = pearsonByType[type].map(item => item.coefficient);
    const avgCorrelation = correlations.reduce((total, val) => total + val,
0) / correlations.length;
    const avgCoefficient = coefficients.reduce((total, val) => total + val,
0) / coefficients.length;
    averagePearsonPerType.push({ type, avgCorrelation, avgCoefficient });
};
```

```
console.log('\nPearson Correlations and Coefficients for Each Result:');
console.table(pearsonResults);

console.log('\nAverage Pearson Correlation and Coefficient by Type:');
console.table(averagePearsonPerType);

console.log('\nAveraged Big Five Responses for Each Type:');
console.table(typeAverages);

// ——————————————————————————————————————————
// Extended Outputs Based on Type Averages
// ——————————————————————————————————————————

/**
 * Compute a cluster average based on a list of types and the pre-computed
typeAverages.
 * @param {Array<string>} clusterTypes - List of personality types in the
cluster.
 * @param {Object} typeAverages - The object holding average responses per
type.
 * @returns {Object} - An object with the cluster average for each trait.
 */
function computeClusterAverage(clusterTypes, typeAverages) {
  const cluster = {};
  traits.forEach(trait => {
    let sum = 0;
    let count = 0;
    clusterTypes.forEach(type => {
      if (typeAverages[type]) {
        sum += typeAverages[type][trait];
        count++;
      }
    });
    cluster[trait] = count > 0 ? sum / count : 0;
  });
  return cluster;
}
```

```
/**
 * Helper to compute cluster average based on a filtering function applied to
the type names.
 * @param {Function} filterFn - A function that receives a type and returns
true if it belongs to the cluster.
 * @param {Object} typeAverages - The object holding average responses per
type.
 * @returns {Object} - An object with the cluster average for each trait.
 */
function computeClusterAverageFromTypeAverages(filterFn, typeAverages) {
    const selectedTypes = Object.keys(typeAverages).filter(filterFn);
    return computeClusterAverage(selectedTypes, typeAverages);
}

// 1. Averages for SF/NT vs ST/NF (using type string comparisons)
const avgSFNTFSTN = computeClusterAverageFromTypeAverages(
  type => type.toLowerCase().includes('sf') ||
type.toLowerCase().includes('nt'),
  typeAverages
);
const avgSTNFTSFN = computeClusterAverageFromTypeAverages(
  type => type.toLowerCase().includes('st') ||
type.toLowerCase().includes('nf'),
  typeAverages
);



// 2. Extended grouping for Ego and Super Ego variants.
const avgSFNT = computeClusterAverageFromTypeAverages(
  type => type === 'ENTP' || type === 'INTJ' || type === 'ISFJ' || type ===
'ESFP',
  typeAverages
);
const avgFSTN = computeClusterAverageFromTypeAverages(
  type => type === 'INTP' || type === 'ENTJ' || type === 'ESFJ' || type ===
'ISFP',
  typeAverages
);
const avgSTNF = computeClusterAverageFromTypeAverages(
  type => type === 'ESTP' || type === 'ISTJ' || type === 'INFJ' || type ===
'ENFP',
  typeAverages
);
const avgTSFN = computeClusterAverageFromTypeAverages(
  type => type === 'ISTP' || type === 'ESTJ' || type === 'ENFJ' || type ===
'INFP',
  typeAverages
);
```

```
// 3. Averages for Introverts vs. Extraverts.
const avgIntroverts = computeClusterAverageFromTypeAverages(
  type => type.startsWith('I'),
  typeAverages
);
const avgExtraverts = computeClusterAverageFromTypeAverages(
  type => type.startsWith('E'),
  typeAverages
);

// 4. Averages for Sensors vs. Intuitors (using type string content).
const avgSensors = computeClusterAverageFromTypeAverages(
    type => type.includes('S'),
    typeAverages
  );
  const avgIntuitors = computeClusterAverageFromTypeAverages(
    type => type.includes('N'),
    typeAverages
  );

// 5. Averages for Thinkers vs. Feelers (using type string content).
const avgThinkers = computeClusterAverageFromTypeAverages(
  type => type.includes('T'),
  typeAverages
);
const avgFeelers = computeClusterAverageFromTypeAverages(
  type => type.includes('F'),
  typeAverages
);

// 6. Averages for Judgers vs. Perceivers (based on type endings)
const avgJudgers = computeClusterAverageFromTypeAverages(
    type => type.endsWith('J'),
    typeAverages
  );
  const avgPerceivers = computeClusterAverageFromTypeAverages(
    type => type.endsWith('P'),
    typeAverages
  );
```

```
// ——————————————————————————————————————
// Save all computed data to a JSON file with tables for each type's
averages.
// ——————————————————————————————————————

console.log('\nAveraged Big Five Responses for Introverts:');
console.table(avgIntroverts);
console.log('\nAveraged Big Five Responses for Extraverts:');
console.table(avgExtraverts);
console.log('\nAveraged Big Five Responses for Sensors:');
console.table(avgSensors);
console.log('\nAveraged Big Five Responses for Intuitors:');
console.table(avgIntuitors);
console.log('\nAveraged Big Five Responses for Thinkers:');
console.table(avgThinkers);
console.log('\nAveraged Big Five Responses for Feelers:');
console.table(avgFeelers);
console.log('\nAveraged Big Five Responses for Judgers:');
console.table(avgJudgers);
console.log('\nAveraged Big Five Responses for Perceivers:');
console.table(avgPerceivers);
console.log('\nAveraged Big Five Responses for SF/NT and FS/TN:');
console.table(avgSFNTFSTN);
console.log('\nAveraged Big Five Responses for ST/NF and TS/FN:');
console.table(avgSTNFTSFN);
console.log('\nAveraged Big Five Responses for SF/NTs:');
console.table(avgSFNT);
console.log('\nAveraged Big Five Responses for FS/TNs:');
console.table(avgFSTN);
console.log('\nAveraged Big Five Responses for ST/NFs:');
console.table(avgSTNF);
console.log('\nAveraged Big Five Responses for TS/FNs:');
console.table(avgTSFN);

const output = {
    typeAverages,
    pearsonResults,
    averagePearsonPerType,
    avgSFNTFSTN,
    avgSTNFTSFN,
    avgJudgers,
    avgPerceivers,
    avgSFNT,
    avgFSTN,
    avgSTNF,
    avgTSFN,
    avgIntroverts,
    avgExtraverts,
    avgThinkers,
    avgFeelers,
    avgSensors,
    avgIntuitors
};

const outputFilePath = path.join(__dirname, 'output.json');
fs.writeFileSync(outputFilePath, JSON.stringify(output, null, 2));
console.log(`\nAll output data has been saved to ${outputFilePath}`);
```